EXPERIMENT 15

AIM:

To understand the importance of data preprocessing and handle missing values in a dataset.

CODE:

```
import numpy as np

import pandas as pd


# Load the dataset

df = pd.read_csv("/content/customer_purchase_dataset.csv")

print(df)


# Display dataset information

df.info()


# Find the mode of City

print(df.City.mode())

print(df.City.mode()[0])

print(type(df.City.mode()))


# Handle missing values in City

df.City.fillna(df.City.mode()[0], inplace=True)


# Handle missing values in Age using median

df.Age.fillna(df.Age.median(), inplace=True)


# Handle missing values in Income using mean
```

```
df.Income.fillna(round(df.Income.mean()), inplace=True)

# Display the dataset after handling missing values
print(df)

# Encode City using one-hot encoding
print(pd.get_dummies(df.City))

# Combine encoded City with selected columns
updated_dataset = pd.concat(
    [pd.get_dummies(df.City), df.iloc[:, [1, 2, 4, 5]]],
    axis=1
)

print(updated_dataset)

# Display updated dataset information
updated_dataset.info()

# Convert Purchased values into numerical values
updated_dataset.Purchased.replace(
    ["No", "Yes"], [0, 1], inplace=True
)

# Display the final preprocessed dataset
print(updated_dataset)
```

OUTPUT:

1. Original Customer Purchase Dataset

Index	Customer_ID	Gender	Age	City	Income	Purchased
0	301	Male	22.0	Chennai	25000.0	No
1	302	Female	25.0	Bangalore	40000.0	Yes
2	303	Male	28.0	Chennai	52000.0	Yes
3	304	Female	24.0	Hyderabad	NaN	No
4	305	Male	30.0	Bangalore	65000.0	Yes
5	306	Female	27.0	Chennai	48000.0	Yes
6	307	Male	NaN	Hyderabad	35000.0	No
7	308	Female	29.0	Bangalore	NaN	Yes
8	309	Male	32.0	Chennai	72000.0	Yes
9	310	Female	26.0	Hyderabad	42000.0	No

2. Dataset Information (df.info())

RangeIndex: 10 entries, 0 to 9
Data columns (total 6 columns):
Customer_ID 10 non-null int64
Gender 10 non-null object
Age 9 non-null float64
City 10 non-null object
Income 8 non-null float64
Purchased 10 non-null object
dtypes: float64(2), int64(1), object(3)
memory usage: 612.0+ bytes

3. Finding and Filling Missing Values

Column	Method	Value Used
City	Mode	Chennai
Age	Median	27.0
Income	Rounded mean	47375

Mode output: df.City.mode() returns Chennai. The notebook fills missing City values with the mode, missing Age with the median, and missing Income with the rounded mean.

4. Dataset After Filling Missing Values

Index	Customer_ID	Gender	Age	City	Income	Purchased
0	301	Male	22.0	Chennai	25000.0	No
1	302	Female	25.0	Bangalore	40000.0	Yes
2	303	Male	28.0	Chennai	52000.0	Yes
3	304	Female	24.0	Hyderabad	47375.0	No
4	305	Male	30.0	Bangalore	65000.0	Yes
5	306	Female	27.0	Chennai	48000.0	Yes
6	307	Male	27.0	Hyderabad	35000.0	No
7	308	Female	29.0	Bangalore	47375.0	Yes
8	309	Male	32.0	Chennai	72000.0	Yes
9	310	Female	26.0	Hyderabad	42000.0	No

5. One-Hot Encoding of City

Output of pd.get_dummies(df.City):

Index	Bangalore	Chennai	Hyderabad
0	False	True	False
1	True	False	False
2	False	True	False
3	False	False	True
4	True	False	False
5	False	True	False
6	False	False	True
7	True	False	False
8	False	True	False
9	False	False	True

6. Updated Dataset After One-Hot Encoding

Idx	Banga-lore	Chennai	Hydera-bad	Gender	Age	Income	Purchased
0	False	True	False	Male	22.0	25000.0	No
1	True	False	False	Female	25.0	40000.0	Yes
2	False	True	False	Male	28.0	52000.0	Yes
3	False	False	True	Female	24.0	47375.0	No
4	True	False	False	Male	30.0	65000.0	Yes
5	False	True	False	Female	27.0	48000.0	Yes
6	False	False	True	Male	27.0	35000.0	No
7	True	False	False	Female	29.0	47375.0	Yes
8	False	True	False	Male	32.0	72000.0	Yes
9	False	False	True	Female	26.0	42000.0	No

7. Updated Dataset Information

RangeIndex: 10 entries, 0 to 9
Data columns (total 7 columns):
Bangalore 10 non-null bool
Chennai 10 non-null bool
Hyderabad 10 non-null bool
Gender 10 non-null object
Age 10 non-null float64
Income 10 non-null float64
Purchased 10 non-null object
dtypes: bool(3), float64(2), object(2)
memory usage: 482.0+ bytes

8. Encoding Purchased as 0 and 1

The notebook replaces 'No' with 0 and 'Yes' with 1.

Idx	Banga-lore	Chennai	Hyder bad	Gender	Age	Income	Purchased
0	False	True	False	Male	22.0	25000.0	0
1	True	False	False	Female	25.0	40000.0	1
2	False	True	False	Male	28.0	52000.0	1
3	False	False	True	Female	24.0	47375.0	0
4	True	False	False	Male	30.0	65000.0	1
5	False	True	False	Female	27.0	48000.0	1
6	False	False	True	Male	27.0	35000.0	0
7	True	False	False	Female	29.0	47375.0	1
8	False	True	False	Male	32.0	72000.0	1
9	False	False	True	Female	26.0	42000.0	0

RESULT:

The given dataset was successfully preprocessed by handling missing values and preparing it for further analysis.